

TEXT MINING USING WEKA PRACTICAL NO 7

@relation movie_reviews

@attribute review string

@attribute sentiment {Positive,Negative}

@data

'Movie was excellent and acting was superb',Positive

'Very boring movie and poor story',Negative

'Amazing direction and fantastic music',Positive

'Waste of time and horrible acting',Negative

'Excellent screenplay and wonderful performance',Positive

'Bad editing and weak storyline',Negative

'Superb cinematography and nice songs',Positive

'Terrible movie and disappointing ending',Negative

'Brilliant acting and very emotional film',Positive

'Worst movie ever watched',Negative

Procedure

Step 1: Open WEKA

Procedure

1. Launch WEKA software.
2. WEKA GUI Chooser appears.
3. Click:

Explorer

What this step does

Opens WEKA Explorer environment for:

- Preprocessing
- Classification
- Text analysis

Step 2: Load Dataset

Procedure

1. Open:

Preprocess Tab

2. Click:

Open File

3. Select:

movie_reviews.arff

What this step does

WEKA:

- Loads movie review dataset
- Reads text records
- Displays attributes

What You Observe

Information	Meaning
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Relation Name	movie_reviews
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Attributes	review, sentiment
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Instances	10
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Why this step is important

Without loading:

- WEKA cannot process text data.

Step 3: Analyze Dataset

Procedure

Click attributes:

- review
- sentiment

Observe:

- String attribute
 - Class distribution
-

What this step does

Helps understand:

- Dataset structure
 - Positive vs Negative review count
-

Step 4: Convert Text to Numerical Data

Why conversion is necessary?

Machine learning algorithms:

- Cannot understand raw text directly.

WEKA converts:

words → numerical vectors

Procedure

Step 1: Choose Filter

1. Click:

Choose

2. Select:

unsupervised → attribute → StringToWordVector

What this filter does

This is the MOST IMPORTANT step.

It converts:

- Sentences into word vectors.
-

Example

Sentence:

Excellent movie with good acting

becomes:

excellent movie good acting

1 1 1 1

Why this is important

Machine learning algorithms work only on:

- Numeric data.

Step 5: Configure StringToWordVector Filter

Click filter name.

Set options like:

Option	Purpose
TF-IDF	Word importance
LowerCaseTokens	Converts to lowercase
OutputWordCounts	Counts word frequency

What each option does

1. LowerCaseTokens

Converts:

Excellent → excellent

Prevents duplicate word meanings.

2. Stopword Removal

Removes common words like:

the, is, was, and

Why?

These words:

- Do not help classification.

3. TF-IDF

TF-IDF means:

Term Frequency – Inverse Document Frequency

Purpose of TF-IDF

Measures:

- Importance of words.

Words appearing frequently:

- Get higher importance.
-

Step 6: Apply Filter

Procedure

Click:

Apply

What this step does

WEKA now:

- Converts text into numeric attributes.
-

What You Observe

Before transformation:

review

Movie was excellent

After transformation:

excellent movie acting boring

1 1 1 0

Why this transformation is important

Makes text:

- Machine-readable
 - Suitable for classification
-

Step 7: Set Class Attribute

Procedure

Select:

sentiment

as class attribute.

What this step does

Defines:

- Target variable for prediction.

Possible classes:

Positive

Negative

Step 8: Open Classify Tab

Procedure

Click:

Classify

What this step does

Opens machine learning classification environment.

Step 9: Choose Classifier

Procedure

Click:

Choose

Select:

bayes → NaiveBayes

What this step does

Naive Bayes:

- Learns patterns from reviews.
 - Predicts sentiment class.
-

Why Naive Bayes is used?

Naive Bayes works very well for:

- Text classification
 - Sentiment analysis
 - Spam filtering
-

Step 10: Set Test Option

Choose:

10-Fold Cross Validation

What this step does

Dataset is divided into:

- Training data
- Testing data

This gives:

- Better accuracy evaluation.
-

Step 11: Click Start

Procedure

Press:

Start

What this step does

WEKA:

- Trains classifier
 - Predicts sentiments
 - Calculates accuracy
-

5. Understanding Output

WEKA displays:

Output	Meaning
Accuracy	Correct predictions
Precision	Exactness
Recall	Completeness
F-Measure	Combined metric
Confusion Matrix	Error analysis

Example Accuracy

Correctly Classified Instances = 90%

Meaning:

- 90% reviews classified correctly.

Step 5: Configure StringToWordVector Filter in WEKA

This is the most important step in Experiment 7 because WEKA converts text reviews into numerical word vectors here.

Goal of Step 5

We want WEKA to:

- Break sentences into words
 - Remove unnecessary words
 - Convert text into machine-readable numbers
-

Before Step 5

You should already have:

- Opened WEKA
- Loaded:

movie_reviews.arff

- Selected:

StringToWordVector

from:

Choose → unsupervised → attribute → StringToWordVector

How to Open Configuration Window

Procedure

After selecting:

StringToWordVector

you will see something like:

`weka.filters.unsupervised.attribute.StringToWordVector`

beside the filter section.

Step

Click directly on:

StringToWordVector

(text beside the filter).

What Happens?

A configuration/settings window opens.

This window contains:

- Tokenizer settings
 - TF-IDF options
 - Stopword settings
 - Word frequency settings
-

Important Settings to Configure

1. LowerCaseTokens

What it does

Converts all words into lowercase.

Example:

Original	Converted
----------	-----------

Excellent	excellent
-----------	-----------

Original Converted

MOVIE movie

Why Important?

Without lowercase conversion:

Movie

movie

MOVIE

would be treated as:

- 3 different words.
-

How to Enable

Find:

LowerCaseTokens

Set:

True

2. OutputWordCounts

What it does

Counts how many times a word appears.

Example:

Sentence:

good good movie

Output:

Word Count

good 2

movie 1

Why Important?

Word frequency helps:

- Sentiment prediction

- Classification accuracy

How to Enable

Find:

OutputWordCounts

Set:

True

3. TF-IDF Transformation

What is TF-IDF?

TF-IDF gives importance to useful words.

Formula:

$$TF\backslash mbox-IDF=TF\times IDF$$

Meaning

Term Meaning

TF Term Frequency

IDF Inverse Document Frequency

Example

Word:

excellent

appears in many positive reviews.

It gets:

- High importance.

Why Important?

TF-IDF:

- Improves classification accuracy.
- Reduces effect of common words.

How to Enable

Find:

TFTransform

Set:

True

Then find:

IDFTransform

Set:

True

4. Stopwords Removal

What are Stopwords?

Common words like:

the

is

was

and

Why Remove Them?

These words:

- Do not help classification.
 - Increase unnecessary dimensions.
-

Example

Sentence:

The movie was excellent

After stopword removal:

movie excellent

How to Enable

Find:

StopwordsHandler

Select:

Rainbow

or default stopwords handler.

5. WordsToKeep

What it does

Limits vocabulary size.

Example:

1000

means:

- Keep only top 1000 important words.
-

Why Important?

Without limit:

- Too many words create memory issues.
-

Recommended Value

Set:

1000

6. Tokenizer

What is Tokenizer?

Tokenizer breaks sentences into words.

Example:

Sentence:

Movie was excellent

becomes:

Movie

was

excellent

Default Tokenizer

Use:

WordTokenizer

Recommended Final Settings

Option	Value
LowerCaseTokens	True
OutputWordCounts	True
TFTransform	True
IDFTransform	True
WordsToKeep	1000
Tokenizer	WordTokenizer

After Configuring

Click:

OK

Next Step

Now click:

Apply